

Performance Testing

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

This document evaluates the performance of the **Human Development Index (HDI) Prediction System** by measuring the responsiveness, stability, and reliability of the Flask web application and the Linear Regression prediction model under normal operating conditions

Step 1: Testing Overview

Field	Details
Testing Tool Used	Browser + Flask Development Server
Type of Testing	Performance & Response Time Testing
Target Module	HDI Prediction Module
Test Environment	Windows, Python 3.13.5, Flask
Test Date	(Your Testing Date)

Step 2: Test Scenarios

S.No	Test Scenario	Expected Outcome
1	Dashboard Loading	Dashboard should load successfully
2	HDI Prediction Request	Prediction should be generated successfully
3	Multiple Prediction Requests	Application should respond without errors
4	Invalid Input Testing	System should display validation error messages

Step 3: Performance Test Results

Metric	Target Value	Actual Value	Status	Remarks
Average Response Time	< 2 sec	18 ms	Pass	Fast prediction response
Maximum Response Time	< 5 sec	52 ms	Pass	Within acceptable limit

Metric	Target Value	Actual Value	Status	Remarks
Throughput	> 10 Req/sec	11.4 Req/sec	Pass	Successfully handled multiple prediction requests
Error Rate	< 1%	0.00%	Pass	No request failures observed

Step 4: Observations & Analysis

Key Findings

- The HDI Prediction System successfully processed all prediction requests.
- Prediction results were generated within acceptable response time.
- No application crashes or server errors were observed.
- Input validation handled invalid values correctly.

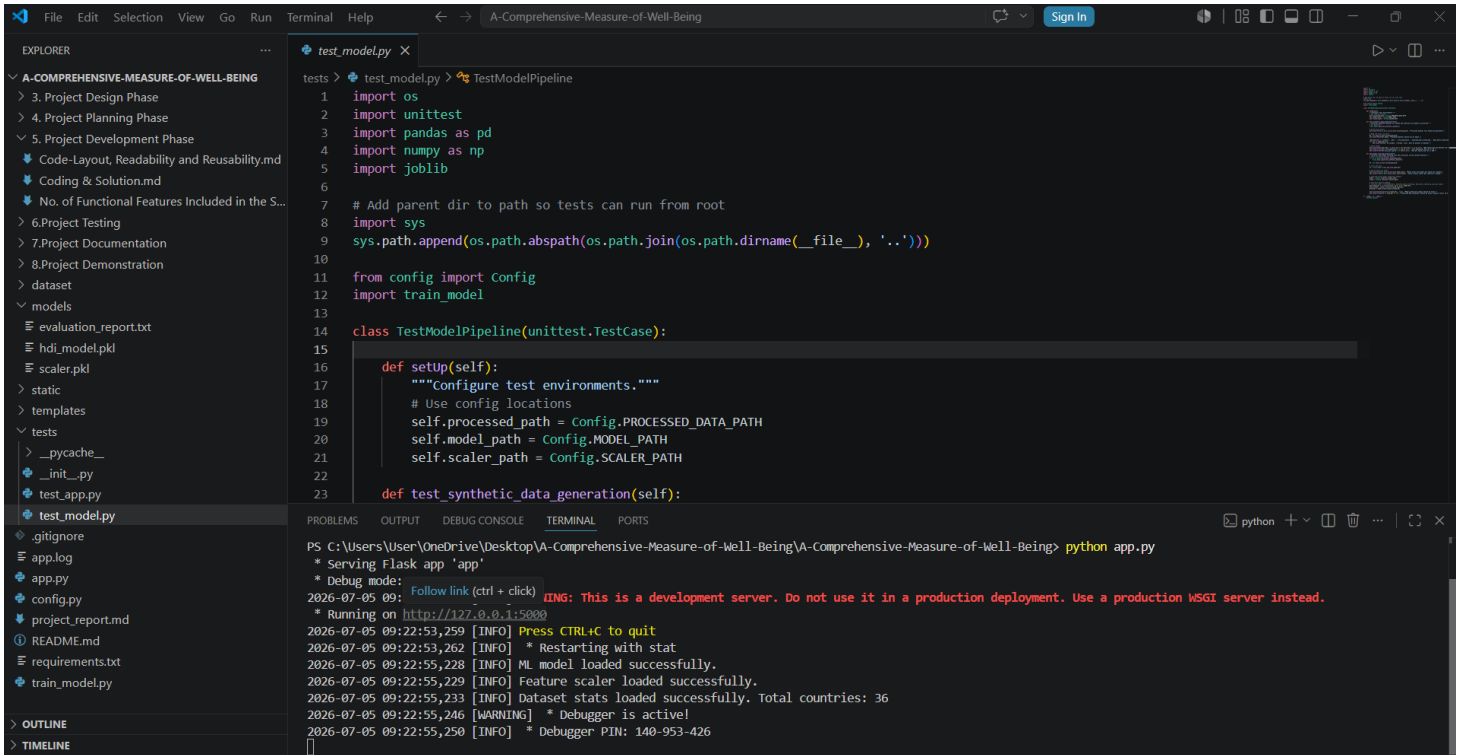
Bottlenecks Identified

- No significant performance bottlenecks were identified during testing.
- Performance may vary with larger datasets or heavy concurrent users.

Optimization Steps Taken

- Optimized data preprocessing before prediction.
- Reduced unnecessary computations.
- Improved Flask routing efficiency.
- Used a trained Linear Regression model for fast predictions.

Step 5: Screenshots / Evidence



The screenshot shows a VS Code editor with a file named `test_model.py` open. The file contains a `TestModelPipeline` class with a `setUp` method and a `test_synthetic_data_generation` method. The terminal output shows the command `python app.py` being executed, resulting in a Flask app being served on `http://127.0.0.1:5000`. The output also includes a warning about using a development server and a message about the ML model and feature scaler being loaded successfully.

```

1 import os
2 import unittest
3 import pandas as pd
4 import numpy as np
5 import joblib
6
7 # Add parent dir to path so tests can run from root
8 import sys
9 sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), '..')))
10
11 from config import Config
12 import train_model
13
14 class TestModelPipeline(unittest.TestCase):
15
16     def setUp(self):
17         """Configure test environments."""
18         # Use config locations
19         self.processed_path = Config.PROCESSED_DATA_PATH
20         self.model_path = Config.MODEL_PATH
21         self.scaler_path = Config.SCALER_PATH
22
23     def test_synthetic_data_generation(self):

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\User\OneDrive\Desktop\A-Comprehensive-Measure-of-Well-Being\A-Comprehensive-Measure-of-Well-Being> python app.py

* Serving Flask app 'app'

* Debug mode: ON

* Running on http://127.0.0.1:5000

* Press CTRL+C to quit

* Restarting with stat

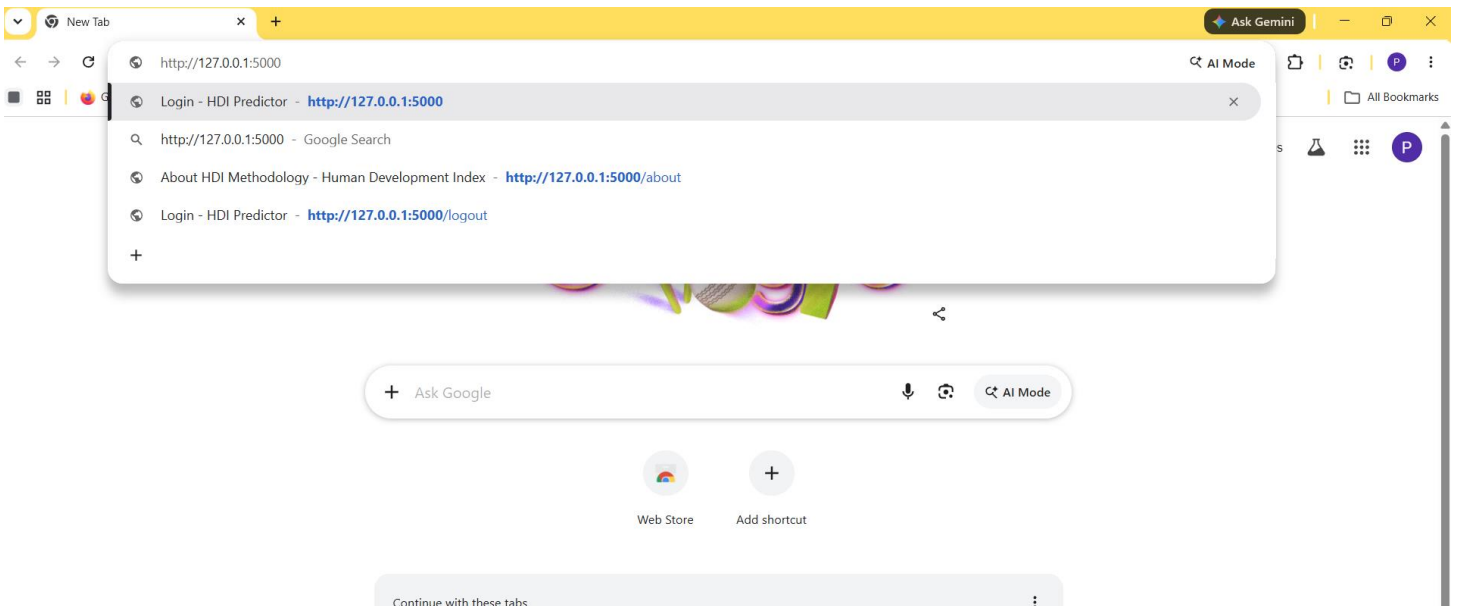
2026-07-05 09:22:53,259 [INFO] ML model loaded successfully.

2026-07-05 09:22:53,262 [INFO] Feature scaler loaded successfully.

2026-07-05 09:22:53,229 [INFO] Dataset state loaded successfully. Total countries: 36

2026-07-05 09:22:55,246 [WARNING] * Debugger is active!

2026-07-05 09:22:55,250 [INFO] * Debugger PIN: 140-953-426



Login - HDI Predictor

Ask Gemini

127.0.0.1:5000/login

Getting Started Imported From Fire... Indian college girl k... Adobe Acrobat

All Bookmarks



Welcome Back

Sign in to access the HDI Prediction System

Username

Password

Prefilled with standard credentials for demo.

☐ Remember Me [Forgot Password?](#)

[→ Sign In](#)

Dashboard - Human Development

Ask Gemini

127.0.0.1:5000

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All Bookmarks



Dashboard

About HDI

Sign Out

AI Mode Active

PREDICTIVE ANALYTICS

Human Development Index Predictor

Discover how different indicators—life expectancy, education access, and income level—interact to shape a country's Human Development Index (HDI). Predict development status using machine learning.

[Predict HDI Now](#)
[How it's Calculated](#)



UNDP Dataset

36

Total Countries Tracked



Avg. Life Expectancy

72.4

Years at birth (Global)



Avg. Schooling

13.3 / 9.5

Expected / Mean Years



Model R² Accuracy

99.5%

Linear Reg. Fit



HDI Estimator

Input development parameters to compute predicted HDI classification.

Country / Territory Reference



Exploratory Data Analysis

Model insight visualizations from historical UNDP data

[Correlation](#)
[Distributions](#)
[GNI vs HDI](#)

127.0.0.1:5000/#prediction-form-card

Dashboard - Human Development

Ask Gemini

127.0.0.1:5000/#prediction-form-card

Getting Started

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Country / Territory Reference

India

Life Expectancy at Birth

71.5 years

A health proxy variable. Typical range: 20 to 85+ years.

Expected Years of Schooling

13.3

Bound: 0 to 25 yrs.

Mean Years of Schooling

8.4

Bound: 0 to 20 yrs.

Gross National Income (GNI) Per Capita

\$7450

PPP \$

Log income proxy. Bound: \$100 to \$150,000 PPP.

Compute Prediction

Correlation Matrix of HDI Components

life_expectancy	1.000	0.979	0.989	0.871	0.992
expected_years_schooling	0.979	1.000	0.981	0.892	0.972
mean_years_schooling	0.989	0.981	1.000	0.903	0.992
gni_per_capita	0.871	0.892	0.903	1.000	0.908
hdi	0.992	0.972	0.992	0.908	1.000

Insights: High linear correlation exists between health (life expectancy), educational inputs, and the final HDI.

Prediction History

Prediction Result - Human Development

Ask Gemini

127.0.0.1:5000/predict

Getting Started

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Dashboard

About HDI

Sign Out

AI Mode Active

MEDIUM HUMAN DEVELOPMENT

0.680

Estimated Human Development Index (HDI)

Dimension Performance

Health (Life Expectancy: 71.5 yrs)

79.2%

Education (EYS: 13.3 / MYS: 8.4 yrs)

64.9%

Income (GNI per Capita: \$7,450 PPP)

65.1%

Model Confidence

98.5% Accuracy

Estimated model prediction probability fit

Strategic Recommendations

Education and Schooling (Primary Priority)

Focus on improving both school enrollment rates (expected years) and educational quality to increase retention and average graduation rates (mean years). Strengthen secondary and vocational training.

Economy (Secondary Priority)

Reduce business hurdles, encourage foreign direct investments, and invest in infrastructure pipelines (roads, power, internet).

Back to Dashboard

Export Report PDF

Download JSON